



CS 134 Project

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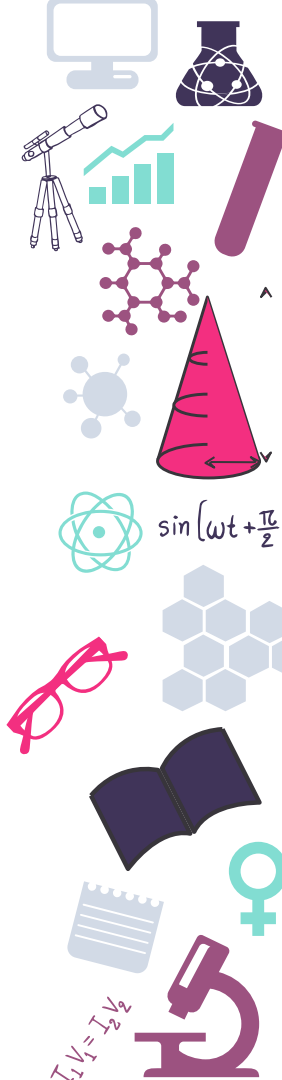


$$\sin\left(\omega t + \frac{\pi}{2}\right)$$

Tangrams

Show a full-color picture of configuration of tangrams - identify the pieces that we need and recreate the picture

- ❖ Variation I: Human interaction - user gets to create their own picture - robot recreates it
- ❖ Variation II: Show a black and white picture of tangrams - algorithm to solve for the configuration - identify the pieces that we need and recreate the picture
- ❖ Variation III: Step by step solving with human



Robot Modes

Recovery Modes:

- ❖ If pieces are not placed properly, or if it was moved, robot should be able to detect and correct mistake
- ❖ If pieces fell, robot should be able to re-pick up the piece
- ❖ If robot cannot find a piece, try to replan

Play Mode:

- ❖ Determine which pieces are needed and pick them up and place them in the right spot
- ❖ Able to flip pieces if necessary

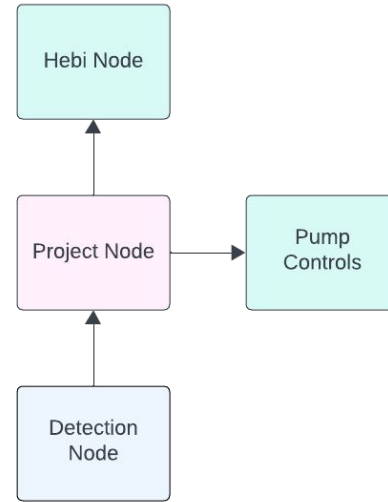
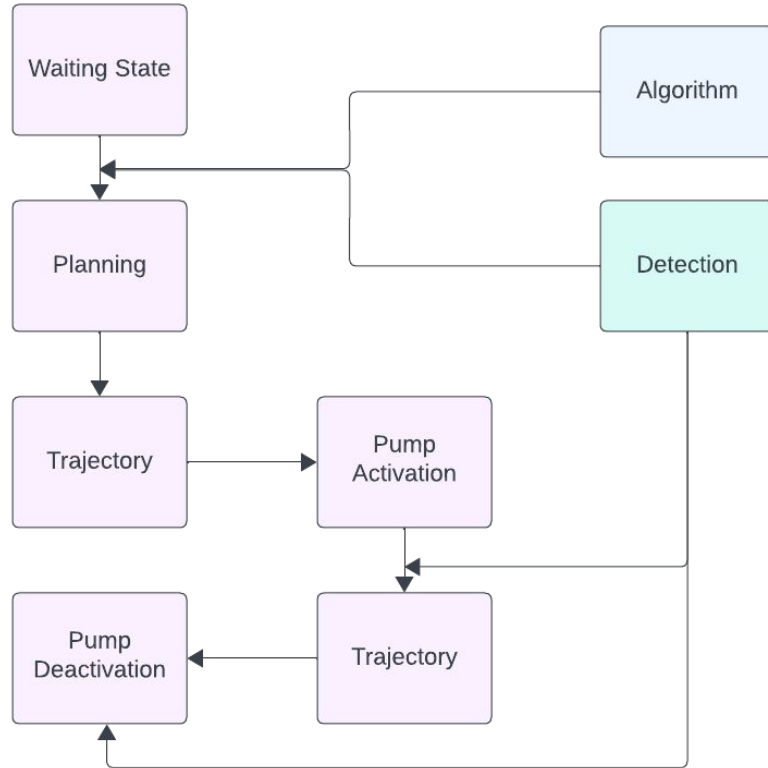
Human Interaction Modes:

- ❖ Step back to let human “play” and then continue once human is done

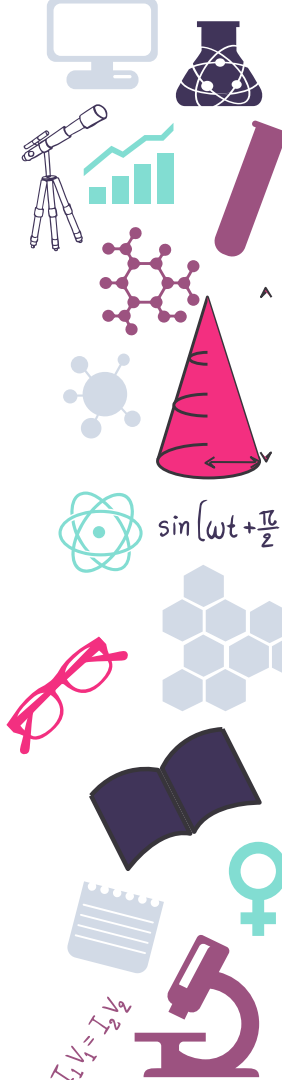
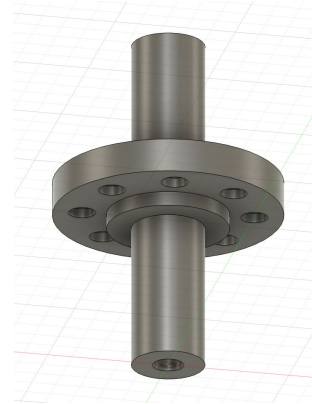
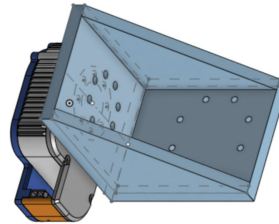
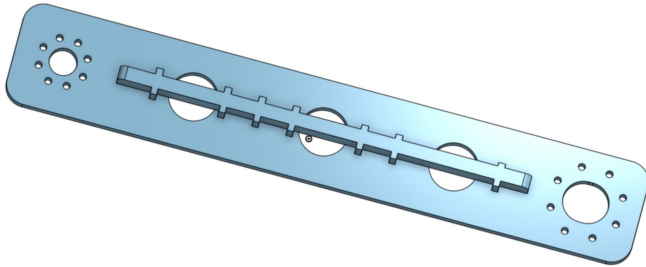
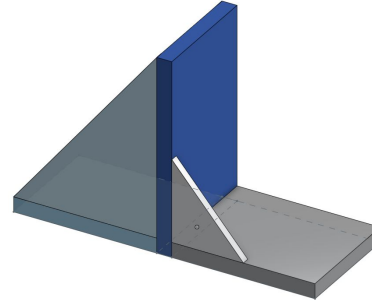
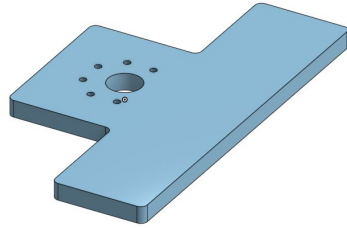
Implications on Robot

- ❖ 5 DOF arm , approx 1m long
 - 3 DOF configuration with 2 DOF wrist
 - 1 motor to keep pieces parallel to the ground and flip pieces
 - 1 motor to rotate pieces to needed configuration
- ❖ ~1m x 1m Workspace
 - Attach robot to middle of the long side of the table
 - Add board/table cover with ArUco tag coordinates on tabletop

Robot Control



Manufactured Parts of Robot



Detection

- ❖ Realsense attached to the ceiling
 - Depth sensing for human interaction
 - Stereo camera for piece detection
- ❖ OpenCV for polygon detection
 - Use different colors for masking
 - Use `cv2.approxPolyDP()` for making polygon contours
- ❖ Contact force detection

Grippers

Suction Cup + Vacuum Pump

- ❖ Use suction cups with $\frac{1}{4}$ " plastic pipes attached to vacuum pump
- ❖ Will allow us to pick up pieces Without modifying them
- ❖ Potentially high risk item



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